

sTMS Interface Functional Specification

Ngee Ann Polytechnic – Timetable

Version 1.0

Document Version

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0.1	12/09/2022	Initial version of NP Timetable IFS	AvePoint
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0.3	21/09/2022	Revised Interface Specification based on NP's comments	AvePoint
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Document Acceptance Form

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Table of Contents

Document Version	1
Document Acceptance Form	2
1. Introduction	4
1.1 Purpose of This Document	4
1.2 Intended Audience	4
2. Technical Overview	5
2.1 System Flow	5
2.2 Interface Design Summary	6
2.2.1 Out-bound Interface	6
2.2.2 In-bound Interface	7
2.3 Business Rule	8
3. Interface Fields	9
3.1 Out-bound Interface Fields	9
3.1.1 COURSE CATALOG	9
3.1.2 COURSE OFFER	10
3.1.3 COURSE COMPONENT	12
3.1.4 COURSE ATTRIBUTES	13
3.1.5 COURSE OFFER OWNER	14
3.1.6 COURSE ATTENDENCE	14
3.1.7 CLASS TIMETABLING	15
3.1.8 CLASS ATTRIBUTE	20
3.1.9 CLASS MTG PAT	20
3.1.10 CLASS COMPONENT	21
3.1.11 CLASS ASSOCIATION	22
3.1.12 A_ENR_CLAS_SSET	23
3.2 In-bound Interface Fields	23
3.2.1 FACILITY	23
4. Assumption and Constraints	25
4.1 Assumption	25
4.2 Constraints	25
5. Appendix	26
5.1 STRM Rule	26
5.2 ACAD_CAREER Mapping Rule	27

1. Introduction

1.1 Purpose of This Document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specifications between the SaaS shared Training Management System (sTMS) and Ngee Ann Polytechnic (NP) NPAL System for transferring sTMS Timetable data and class session facility data.

sTMS shall identify all NP Timetable data with changes, generate corresponding CSV files for these data, deposit CSV files encrypted by PGP protocol into sTMS SFTP server accordingly. Scheduled job to be implemented by NP will pull and consume the data files accordingly for data structure setup in NPAL System.

NP shall generate CSV file for class meeting facility data, deposit the CSV files encrypted by PGP protocol into sTMS SFTP server, and sTMS will update facility for course intake timetable accordingly.

1.2 Intended Audience

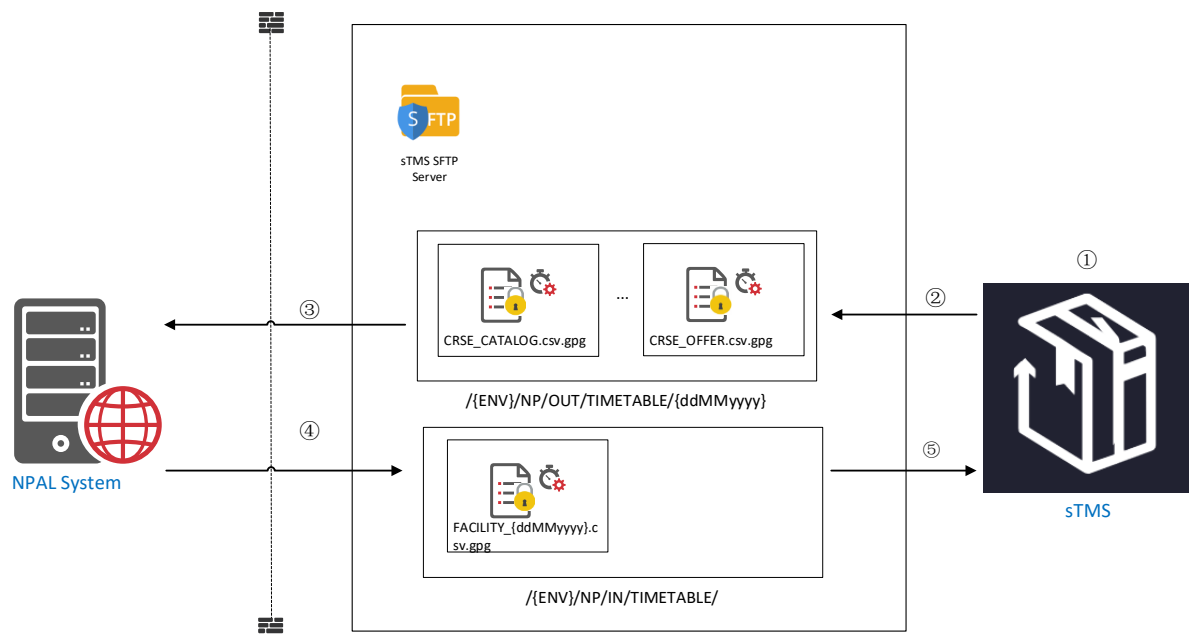
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of shared Training Management System and NPAL System.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **NPAL System:** Ngee Ann Polytechnic NPAL system.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted files used for data exchange.
- **sTMS:** shared Training Management System.



*The above system flow only shows some CSV files encryption with PGP protocol. For the interface file name of all NP timetable data CSV files, please refer to [2.2 Interface Design Summary](#).

SN	Description
1	sTMS generates NP timetable data CSV files based on institution, table name, modified time on a pre-defined schedule. For file generation rules can refer to 2.3. Business Rule
2	sTMS configures timer job to generate folders for different kind of resources: <code>/ENV/NP/OUT/TIMETABLE/</code> in the sTMS SFTP server daily.

	sTMS pushes NP timetable data CSV files under {ddMMyyyy} folder after encryption with PGP protocol.
3	NP pulls these CSV files from sTMS SFTP server on demand.
4	NP generate the whole list of CET running and upcoming classes (the historical classes will not be included) with Facility ids by CSV file and deposits to sTMS SFTP server after encryption with PGP protocol.
5	sTMS pulls and decrypts Facility CSV file and perform a comparison, then update those different session facility ids in sTMS.

2.2 Interface Design Summary

2.2.1 Out-bound Interface

Interface File Name	COURSE_CATALOG_{ddMMyyyy}.csv.gpg COURSE_OFFER_{ddMMyyyy}.csv.gpg COURSE_COMPONENT_{ddMMyyyy}.csv.gpg COURSE_ATTRIBUTES_{ddMMyyyy}.csv.gpg COURSE_OFFER_OWNER_{ddMMyyyy}.csv.gpg COURSE_ATTENDENCE_{ddMMyyyy}.csv.gpg CLASS_TIMETABLING_{ddMMyyyy}.csv.gpg CLASS_ATTRIBUTE_{ddMMyyyy}.csv.gpg CLASS_MTG_PAT_{ddMMyyyy}.csv.gpg CLASS_COMPONENT_{ddMMyyyy}.csv.gpg CLASS_ASSOCIATION_{ddMMyyyy}.csv.gpg A_ENR_CLAS_SSET_{ddMMyyyy}.csv.gpg
Data Source System	Shared Training Management SystemShared Training Management System
Data Destination System	NPAL system
Method	SFTP
File Path	/{ENV}/NP/OUT/TIMETABLE/ ENV: UAT, PROD
Encryption Method	PGP

Frequency	Daily Sync
Timing	01:00 am (TBC)
Folder Retention Policy	Folders under the /{ENV}/NP/OUT/TIMETABLE are retained for a maximum of 30 days in the sTMS SFTP server, after this period the folders will be automatically deleted. NP shall delete these folders after pulling out the files
File Transfer Policy	No file shall be deposited into sTMS SFTP server if there are no new records are added/updated in the table
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer files could not be found in the newly created SFTP folder. ○ AP redeposits the files in the newly created corresponding SFTP folder.

2.2.2 In-bound Interface

Interface File Name	FACILITY_{ddMMyyyy}.csv.gpg
Data Source System	NPAL System
Data Destination System	shared Training Management System
Method	SFTP
File Path	/{ENV}/NP/IN/TIMETABLE/ ENV: UAT, PROD
Encryption Method	PGP
Frequency	Daily Sync
File Transfer Policy	If no Facility file found in SFTP folder, no records will update into sTMS.
Possible Exception and Exception Handling	• sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure.

2.3 Business Rule

CSV files generation rule

Table record trigger point: First-time full load data sync + Subsequent delta data sync.

Filter condition:

- Institution = NP,
- The data fields in the table have changed,
- Modified datetime later than last successful sync datetime.

3. Interface Fields

The symbol  represents the field is the primary key of the table.

3.1 Out-bound Interface Fields

3.1.1 COURSE CATALOG

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update & For any deletion in sTMS, it shall be mapped to Inactive under Eff_Status
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank. NPAL to generate.
EFFDT	DATE	10	NA	Format: DD/MM/YYYY Default to "01/01/1901" Effective date is only set to 01/01/1901 for new modules. Subsequent changes/updates to modules should update and follow the date format of DD/MM/YYYY
EFF_STATUS	VARCHAR2(1 CHAR)	1	Status	"A": Active "I": Inactive
DESCR	VARCHAR2(30 CHAR)	30	Module name	NIL
EQUIV_CRSE_ID	VARCHAR2(5 CHAR)	5	NA	Default to " "
CONSENT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ALLOW_MULT_ENROLL	VARCHAR2(1 CHAR)	1	NA	Default to "N"
UNITS_MINIMUM	NUMBER(5,2)	3.2	No. of module hours	NIL
UNITS_MAXIMUM	NUMBER(5,2)	3.2	No. of module hours	NIL
UNITS_ACAD_PROG	NUMBER(5,2)	3.2	No. of module hours	NIL

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
UNITS_FINAID_PROG	NUMBER(5,2)	3.2	No. of module hours	NIL
CRSE_REPEATABLE	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
UNITS_REPEAT_LIMIT	NUMBER(5,2)	3.2	NA	Default to "999"
CRSE_REPEAT_LIMIT	NUMBER(38,0)	2	NA	Default to "99"
GRADING_BASIS	VARCHAR2(3 CHAR)	3	NA	Default to "NON"
GRADE_ROSTER_PRINT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
SSR_COMPONENT	VARCHAR2(3 CHAR)	3	NA	Default to "LEC"
COURSE_TITLE_LONG	VARCHAR2(100 CHAR)	100	Module name	NIL
LST_MULT_TRM_CRS	VARCHAR2(1 CHAR)	1	NA	Default to "N"
CRSE_CONTACT_HRS	NUMBER(5,2)	3.2	No. of module hours	NIL
RQMNT_DESIGNATION	VARCHAR2(4 CHAR)	4	NA	Default to " "
CRSE_COUNT	NUMBER(4,2)	2.2	NA	Default to "1"
INSTRUCTOR_EDIT	VARCHAR2(2 CHAR)	2	NA	Default to "I"
FEES_EXIST	VARCHAR2(1 CHAR)	1	NA	Default to "LEC"
COMPONENT_PRIMARY	VARCHAR2(3 CHAR)	3	NA	Default to "1"
ENRL_UN_LD_CLC_TYP	VARCHAR2(1 CHAR)	1	NA	Default to "1"
SSR_DROP_CONSENT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
SCC_ROW_ADD_OWNER	VARCHAR2(30 CHAR)	30	Created By	Sample: Eric Zheng
SCC_ROW_UPD_OWNER	VARCHAR2(30 CHAR)	30	Modified By	Sample: Sam Chia

3.1.2 COURSE OFFER

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
				U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
EFFDT	DATE	10	NA	Format: DD/MM/YYYY Default to "01/01/1901" Effective date is only set to 01/01/1901 for new modules. Subsequent changes/updates to modules should update and follow the date format of DD/MM/YYYY
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to "1"
INSTITUTION	VARCHAR2(5 CHAR)	5	NA	Default to "NPOLY"
ACAD_GROUP	VARCHAR2(5 CHAR)	5	School code	NIL
SUBJECT	VARCHAR2(8 CHAR)	8	Course code	NIL
CATALOG_NBR	VARCHAR2(10 CHAR)	10	Module code	NIL
COURSE_APPROVED	VARCHAR2(1 CHAR)	1	NA	Default to "A"
CAMPUS	VARCHAR2(5 CHAR)	5	NA	Default to "CLEM"
SCHEDULE_PRINT	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
CATALOG_PRINT	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
SCHED_PRINT_INSTR	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
ACAD_ORG	VARCHAR2(10 CHAR)	10	School code	NIL
ACAD_CAREER	VARCHAR2(4 CHAR)	4	Course category + Course type	Refer to ACAD_CAREER Rule in Appendix
SPLIT_OWNER	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
SCHED_TERM_ROLE	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
RQRMNT_GROUP	VARCHAR2(6 CHAR)	6	NA	Set as blank
USE_BLIND_GRADING	VARCHAR2(1 CHAR)	1	NA	Default to "N"

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
RCV_FROM_ITEM_TYPE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
GL_INTERFACE_REQ	VARCHAR2(1 CHAR)	1	NA	Default to "N"
SEL_GROUP	VARCHAR2(10 CHAR)	10	NA	Set as blank
SCHEDULE_COURSE	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
OEE_IND	VARCHAR2(1 CHAR)	1	NA	Default to "N"
SSR_AC_EXAM_ONLY	VARCHAR2(1 CHAR)	1	NA	Default to "N"
SCC_ROW_ADD_OWNER	VARCHAR2(30 CHAR)	30	Created By	Sample: Eric Zheng
SCC_ROW_UPDATE_OWNER	VARCHAR2(30 CHAR)	30	Modified By	Sample: Sam Chia

3.1.3 COURSE COMPONENT

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
EFFDT	DATE	10	NA	Format: DD/MM/YYYY Default to "01/01/1901" Effective date is only set to 01/01/1901 for new modules. Subsequent changes/updates to modules should update and follow the date format of DD/MM/YYYY
SSR_COMPONENT	VARCHAR2(3 CHAR)	3	NA	Default to "LEC"
OPTIONAL_SECTION	VARCHAR2(1 CHAR)	1	NA	Set the value to "N"

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
DEFAULT_SECT_SIZE	NUMBER(38,0)	3	NA	Set the value to 25
CONTACT_HOURS	NUMBER(7,3)	4.3	NA	Can be set to 0
FINAL_EXAM	VARCHAR2(2 CHAR)	2	NA	Default to "N"
EXAM_SEAT_SPACING	NUMBER(38,0)	1	NA	Default to 1
DYN_DT_INCLUDE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
AUTO_CREATE_CMPNT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ATTEND_GENERAL	VARCHAR2(1 CHAR)	1	NA	Default to "N"
WEEK_WORKLOAD_HRS	NUMBER(5,2)	3.2	NA	Default to 0
OEE_WORKLOAD_HRS	NUMBER(5,2)	3.2	NA	Default to 0
A_ATT_COMP_LEVEL	VARCHAR2(1 CHAR)	1	NA	Default to blank

3.1.4 COURSE ATTRIBUTES

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
EFFDT	DATE	10	NA	Format: DD/MM/YYYY Default to "01/01/1901" Effective date is only set to 01/01/1901 for new modules. Subsequent changes/updates to modules should update and follow the date format of DD/MM/YYYY

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 CRSE_ATTR	VARCHAR2(4 CHAR)	4	Tag	NIL
CRSE_ATTR_VALUE	VARCHAR2(10 CHAR)	10	Tag value	NIL
SCC_ROW_ADD_OPRID	VARCHAR2(30 CHAR)	30	Created By	Sample: Eric Zheng
SCC_ROW_UPD_OPRID	VARCHAR2(30 CHAR)	30	Modified By	Sample: Sam Chia

3.1.5 COURSE OFFER OWNER

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
EFFDT	DATE Effective date is only set to 01/01/1901 for new modules. Subsequent changes/updates to modules should update and follow the date format of DD/MM/YYYY	10	NA	Format: DD/MM/YYYY Default to "01/01/1901"
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to 1
ACAD_ORG	VARCHAR2(10 CHAR)	10	School code	NIL
PERCENT_OWNED	NUMBER(6,3)	3.3	NA	Default to 100

3.1.6 COURSE ATTENDENCE

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
EFFDT	DATE	10	NA	Format: DD/MM/YYYY Default to "01/01/1901" Effective date is only set to 01/01/1901 for new modules. Subsequent changes/updates to modules should update and follow the date format of DD/MM/YYYY
SSR_COMPONENT	VARCHAR2(3 CHAR)	3	NA	Default to "LEC"
CLASS_ATTEND_TYPE	VARCHAR2(4 CHAR)	4	NA	Default to "MTG"
INSTRUCTION_MODE	VARCHAR2(2 CHAR)	2	NA	Default to "P"
ATTEND_PRESENT_USE	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
ATTEND_REASON_USE	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
ATTEND_TARDY_USE	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
ATTEND_LEFT_USE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ATTEND_CONTACT_USE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ATTEND_TIME_USE	VARCHAR2(1 CHAR)	1	NA	Default to "Y"
ATTEND_TMPLT_OVRD	VARCHAR2(1 CHAR)	1	NA	Default to "N"

3.1.7 CLASS TIMETABLING

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to 1
🔑 STRM	VARCHAR2(4 CHAR)	4	Course intake semester	Refer to STRM Rule in Appendix
SESSION_CODE	VARCHAR2(3 CHAR)	3	NA	Default to 1
🔑 CLASS_SECTION	VARCHAR2(4 CHAR)	4	Class name	NIL
INSTITUTION	VARCHAR2(5 CHAR)	5	NA	Default to NPOLY
ACAD_GROUP	VARCHAR2(5 CHAR)	5	School code	If multi values detached in sTMS, pick the first one
SUBJECT	VARCHAR2(8 CHAR)	8	Course code	NIL
CATALOG_NBR	VARCHAR2(10 CHAR)	10	Module code	NIL
ACAD_CAREER	VARCHAR2(4 CHAR)	4	Course category + Course type	Refer to ACAD CAREER Rule in Appendix
DESCR	VARCHAR2(30 CHAR)	30	Module Name	NIL
CLASS_NBR	NUMBER(38,0)	5	NA	Set as blank
SSR_COMPONENT	VARCHAR2(3 CHAR)	3	NA	Default to "LEC"
ENRL_STAT	VARCHAR2(1 CHAR)	1	NA	Default to "O"
CLASS_STAT	VARCHAR2(1 CHAR)	1	NA	Default to "A"
CLASS_TYPE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ASSOCIATED_CLASSES	NUMBER(38,0)	4	NA	Default to "1"
WAITLIST_DAEMON	VARCHAR2(2 CHAR)	2	NA	Default to "N"
AUTO_ENRL_WAITLIST	VARCHAR2(1 CHAR)	1	NA	Default to "N"
STDNT_SPEC_PERM	VARCHAR2(1 CHAR)	1	NA	Default to "N"
AUTO_ENROLL_SECT_1	VARCHAR2(4 CHAR)	4	NA	Default to " "
AUTO_ENROLL_SECT_2	VARCHAR2(4 CHAR)	4	NA	Default to " "
RESECTION	VARCHAR2(4 CHAR)	4	NA	Default to " "
SCHEDULE_PRINT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
CONSENT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ENRL_CAP	NUMBER(38,0)	4	NA	Default to 25

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
WAIT_CAP	NUMBER(38,0)	4	NA	Default to "0"
MIN_ENRL	NUMBER(38,0)	4	NA	Default to "0"
ENRL_TOT	NUMBER(38,0)	4	NA	Default to "0"
WAIT_TOT	NUMBER(38,0)	4	NA	Default to "0"
CRS_TOPIC_ID	NUMBER(38,0)	3	NA	Default to "0"
PRINT_TOPIC	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ACAD_ORG	VARCHAR2(10 CHAR)	10	School code	NIL
NEXT_STDNT_POS ITIN	NUMBER(38,0)	8	NA	Default to "10"
EMPLID	VARCHAR2(11 CHAR)	11	NA	Default to " "
CAMPUS	VARCHAR2(5 CHAR)	5	NA	Default to "CLEM"
LOCATION	VARCHAR2(10 CHAR)	10	NA	Default to "CLEMENTI"
CAMPUS_EVENT_ NBR	VARCHAR2(9 CHAR)	9	NA	Default to " "
INSTRUCTION_MO DE	VARCHAR2(2 CHAR)	2	NA	Default to "P"
EQUIV_CRSE_ID	VARCHAR2(5 CHAR)	5	NA	Default to " "
OVRD_CRSE_EQUI V_ID	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ROOM_CAP_REQ UEST	NUMBER(38,0)	4	NA	Default to 25
START_DT	DATE	10	Start date of course intake	NIL
END_DT	DATE	10	End date of course intake	NIL
CANCEL_DT	DATE	10	NA	Default to " "
PRIM_INSTR_SECT	VARCHAR2(4 CHAR)	4	Class name	NIL
COMBINED_SECTI ON	VARCHAR2(1 CHAR)	1	NA	Default to " "
HOLIDAY_SCHEDU LE	VARCHAR2(6 CHAR)	6	NA	Default to "SIN"
EXAM_SEAT_SPAC ING	NUMBER(38,0)	1	NA	Default to "1"
DYN_DT_INCLUDE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
DYN_DT_CALC_RE Q	VARCHAR2(1 CHAR)	1	NA	Default to "N"

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
ATTEND_GENERATE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
ATTEND_SYNC_REQD	VARCHAR2(1 CHAR)	1	NA	Default to "N"
FEES_EXIST	VARCHAR2(1 CHAR)	1	NA	Default to "N"
CNCL_IF_STUDENT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
RCV_FROM_ITEM_TYPE	VARCHAR2(1 CHAR)	1	NA	Default to "N"
AP_BUS_UNIT	VARCHAR2(5 CHAR)	5	NA	Default to " "
AP_LEDGER	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_ACCOUNT	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_DEPTID	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_PROJ_ID	VARCHAR2(15 CHAR)	15	NA	Default to " "
AP_PRODUCT	VARCHAR2(6 CHAR)	6	NA	Default to " "
AP_FUND_CODE	VARCHAR2(5 CHAR)	5	NA	Default to " "
AP_PROG_CODE	VARCHAR2(5 CHAR)	5	NA	Default to " "
AP_CLASS_FLD	VARCHAR2(5 CHAR)	5	NA	Default to " "
AP_AFFILIATE	VARCHAR2(5 CHAR)	5	NA	Default to " "
AP_OP_UNIT	VARCHAR2(8 CHAR)	8	NA	Default to " "
AP_ALTACCT	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_BUD_REF	VARCHAR2(8 CHAR)	8	NA	Default to " "
AP_CF1	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_CF2	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_CF3	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_AFF_INT1	VARCHAR2(10 CHAR)	10	NA	Default to " "
AP_AFF_INT2	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_BUS_UNIT	VARCHAR2(5 CHAR)	5	NA	Default to " "
WRITEOFF_LEDGER	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_ACCOUNT	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_DEPTID	VARCHAR2(10 CHAR)	10	NA	Default to " "

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
WRITEOFF_PROJ_ID	VARCHAR2(15 CHAR)	15	NA	Default to " "
WRITEOFF_PRODUCT	VARCHAR2(6 CHAR)	6	NA	Default to " "
WRITEOFF_FUND_CODE	VARCHAR2(5 CHAR)	5	NA	Default to " "
WRITEOFF_PROG_CODE	VARCHAR2(5 CHAR)	5	NA	Default to " "
WRITEOFF_CLASS_FLD	VARCHAR2(5 CHAR)	5	NA	Default to " "
WRITEOFF_AFFILIATE	VARCHAR2(5 CHAR)	5	NA	Default to " "
WRITEOFF_OP_UNIT	VARCHAR2(8 CHAR)	8	NA	Default to " "
WRITEOFF_ALTACT	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_BUDREF	VARCHAR2(8 CHAR)	8	NA	Default to " "
WRITEOFF_CF1	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_CF2	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_CF3	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_AFF_INT1	VARCHAR2(10 CHAR)	10	NA	Default to " "
WRITEOFF_AFF_INT2	VARCHAR2(10 CHAR)	10	NA	Default to " "
EXT_WRITEOFF	VARCHAR2(50 CHAR)	50	NA	Default to " "
GL_INTERFACE_REQ	VARCHAR2(1 CHAR)	1	NA	Default to "N"
LMS_FILE_TYPE	VARCHAR2(3 CHAR)	3	NA	Default to " "
LMS_GROUP_ID	VARCHAR2(50 CHAR)	50	NA	Default to " "
LMS_URL	VARCHAR2(254 CHAR)	254	NA	Default to " "
LMS_CLASS_EXT_DTTM	TIMESTAMP(6)	26	NA	Default to " "
LMS_ENRL_EXT_DTTM	TIMESTAMP(6)	26	NA	Default to " "
LMS_PROVIDER	VARCHAR2(30 CHAR)	30	NA	Default to " "
SSR_DROP_CONSENT	VARCHAR2(1 CHAR)	1	NA	Default to " "

3.1.8 CLASS ATTRIBUTE

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to 1
 STRM	VARCHAR2(4 CHAR)	4	Course intake semester	Refer to STRM Rule in Appendix
SESSION_CODE	VARCHAR2(3 CHAR)	3	NA	Default to 1
 CLASS_SECTION	VARCHAR2(4 CHAR)	4	Class Name	NIL
 CRSE_ATTR	VARCHAR2(4 CHAR)	4	Module tag	NIL
CRSE_ATTR_VALUE	VARCHAR2(10 CHAR)	10	Module tag value	NIL

3.1.9 CLASS MTG PAT

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to 1
 STRM	VARCHAR2(4 CHAR)	4	Course intake semester	Refer to STRM Rule in Appendix
SESSION_CODE	VARCHAR2(3 CHAR)	3	NA	Default to 1
 CLASS_SECTION	VARCHAR2(4 CHAR)	4	Class name	NIL
 CLASS_MTG_NBR	NUMBER(38,0)	2	To use running number for group recurring session	NIL
FACILITY_ID	VARCHAR2(10 CHAR)	10	Facility Id	NIL
MEETING_TIME_START	TIMESTAMP(6)	15	Recurring sessions start time	NIL

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
MEETING_TIME_END	TIMESTAMP(6)	15	Recurring sessions end time	NIL
MON	VARCHAR2(1 CHAR)	1	Monday	NIL
TUES	VARCHAR2(1 CHAR)	1	Tuesday	NIL
WED	VARCHAR2(1 CHAR)	1	Wednesday	NIL
THURS	VARCHAR2(1 CHAR)	1	Thursday	NIL
FRI	VARCHAR2(1 CHAR)	1	Friday	NIL
SAT	VARCHAR2(1 CHAR)	1	Saturday	NIL
SUN	VARCHAR2(1 CHAR)	1	Sunday	NIL
START_DT	DATE	10	Start date	NIL
END_DT	DATE	10	End date	NIL
CRS_TOPIC_ID	NUMBER(38,0)	3	NA	Default to "0"
DESCR	VARCHAR2(30 CHAR)	30	NA	Default to " "
STND_MTG_PAT	VARCHAR2(4 CHAR)	4	NA	Default to " "
PRINT_TOPIC_ON_XCR	VARCHAR2(1 CHAR)	1	NA	Default to "N"

3.1.10 CLASS COMPONENT

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to 1
 STRM	VARCHAR2(4 CHAR)	4	Course intake semester	Refer to STRM Rule in Appendix
SESSION_CODE	VARCHAR2(3 CHAR)	3	NA	Default to 1
ASSOCIATED_CLASSES	NUMBER(38,0)	4	NA	Default to 1
SSR_COMPONENT	VARCHAR2(3 CHAR)	3	NA	Default to "LEC"
OPTIONAL_SECTION	VARCHAR2(1 CHAR)	1	NA	Default to "N"

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
CONTACT_HOURS	NUMBER(7,3)	4.3	NA	Default to "0"
FINAL_EXAM	VARCHAR2(2 CHAR)	2	NA	Default to "N"
AUTO_CREATE_C MPNT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
WEEK_WORKLOA D_HRS	NUMBER(5,2)	3.2	NA	Default to "0"

3.1.11 CLASS ASSOCIATION

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to "1"
 STRM	VARCHAR2(4 CHAR)	4	Course intake semester	Refer to STRM Rule in Appendix
SESSION_CODE	VARCHAR2(3 CHAR)	3	NA	Default to "1"
ASSOCIATED_CLASSES	NUMBER(38,0)	4	NA	Default to "1"
UNITS_MINIMUM	NUMBER(5,2)	3.2	NA	Default to "0"
UNITS_MAXIMUM	NUMBER(5,2)	3.2	NA	Default to "0"
UNITS_ACAD_PROG	NUMBER(5,2)	3.2	NA	Default to "0"
UNITS_FINAID_PROG	NUMBER(5,2)	3.2	NA	Default to "0"
GRADING_BASIS	VARCHAR2(3 CHAR)	3	NA	Default to "NON"
GRADE_ROSTER_PRINT	VARCHAR2(1 CHAR)	1	NA	Default to "N"
SSR_COMPONENT	VARCHAR2(3 CHAR)	3	NA	Default to "LEC"
BILLING_FACTOR	NUMBER(5,3)	2.3	NA	Default to "1"
CRSE_CONTACT_HOURS	NUMBER(5,2)	3.2	NA	Default to "0"
RQRMNT_GROUP	VARCHAR2(6 CHAR)	6	NA	Default to " "
USE_CATLG_RQS	VARCHAR2(1 CHAR)	1	NA	Default to "Y"

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
RQMNT_DESIGN	VARCHAR2(4 CHAR)	4	NA	Default to " "
CRSE_COUNT	NUMBER(4,2)	2.2	NA	Default to "1"
INSTRUCTOR_EDIT	VARCHAR2(2 CHAR)	2	NA	Default to "N"
USE_BLIND_GRADING	VARCHAR2(1 CHAR)	1	NA	Default to "N"
SEL_GROUP	VARCHAR2(10 CHAR)	10	NA	Default to " "
COMPONENT_PRIMARY	VARCHAR2(3 CHAR)	3	NA	Default to "LEC"
UPDATE_STDNT_ENRL	VARCHAR2(1 CHAR)	1	NA	Default to "N"

3.1.12 A_ENR_CLAS_SSET

NPAL Field Name	Data Type	Length	sTMS Field	Remarks
 INTAKE_NBR	VARCHAR2(20)	20	Course intake number	sTMS course intake number
 MODULE_CODE	VARCHAR2(6)	6	Module code	sTMS module code
ACTION	VARCHAR2(1)	1	NA	C: create new U: update D: delete
CRSE_ID	VARCHAR2(6 CHAR)	6	NA	Set as blank
CRSE_OFFER_NBR	NUMBER(38,0)	2	NA	Default to 1
 STRM	VARCHAR2(4 CHAR)	4	Course intake semester	Refer to STRM Rule in Appendix
SESSION_CODE	VARCHAR2(3 CHAR)	3	NA	Default to 1
 CLASS_SECTION	VARCHAR2(4 CHAR)	4	Class name	NIL
A_STUDENT_SET	VARCHAR2(8 CHAR)	8	NA	Default to " "

3.2 In-bound Interface Fields

3.2.1 FACILITY

NPAL Field Name	Data Type	Length	sTMS Field
CATALOG_NBR	VARCHAR2(10 CHAR)	10	Module code
STRM	VARCHAR2(4 CHAR)	4	Course intake schedule

NPAL Field Name	Data Type	Length	sTMS Field
CLASS_SECTION	VARCHAR2(4 CHAR)	4	Class name
MEETING_TIME_START	TIMESTAMP(6)	15	Start time of class session
MEETING_TIME_END	TIMESTAMP(6)	15	End time of class session
MON	VARCHAR2(1 CHAR)	1	Recurring sessions details
TUES	VARCHAR2(1 CHAR)	1	
WED	VARCHAR2(1 CHAR)	1	
THURS	VARCHAR2(1 CHAR)	1	
FRI	VARCHAR2(1 CHAR)	1	
SAT	VARCHAR2(1 CHAR)	1	
SUN	VARCHAR2(1 CHAR)	1	
START_DT	DATE	15	
END_DT	DATE	15	
CLASS_MTG_NBR	NUMBER(38,0)	2	
FACILITY_ID	VARCHAR2(10 CHAR)	10	Session facility

4. Assumption and Constraints

4.1 Assumption

No.	Assumption

4.2 Constraints

No.	Constraints
1	sTMS will create a new folder called {ddMMyyyy} under /{ENV}/NP/OUT/TIMETABLE/ path daily
2	Datetime values send via csv file default to Singapore Standard Time (SST). (UTC+08:00)

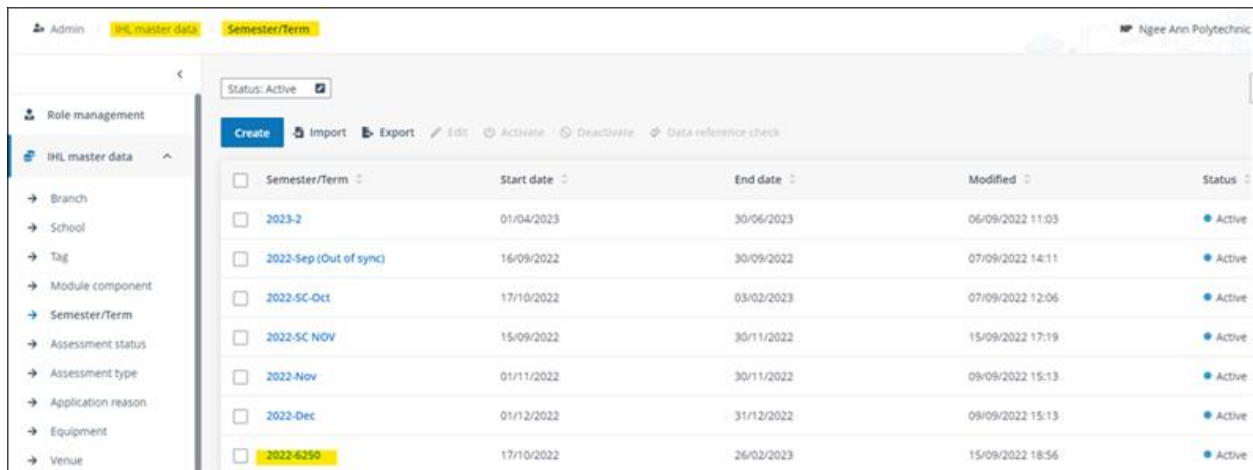
5. Appendix

5.1 STRM Rule

STRM is a 4 digits code following the generation rule:

Full Qualification Courses & Vocation Term Courses:

STRM equals Term field value of Semester in sTMS.



Semester/Term	Start date	End date	Modified	Status
2023-2	01/04/2023	30/06/2023	06/09/2022 11:03	Active
2022-Sep (Out of sync)	16/09/2022	30/09/2022	07/09/2022 14:11	Active
2022-SC-Oct	17/10/2022	03/02/2023	07/09/2022 12:06	Active
2022-SC NOV	15/09/2022	30/11/2022	15/09/2022 17:19	Active
2022-Nov	01/11/2022	30/11/2022	09/09/2022 15:13	Active
2022-Dec	01/12/2022	31/12/2022	09/09/2022 15:13	Active
2022-6250	17/10/2022	26/02/2023	15/09/2022 18:56	Active

Short Courses:

First 2 digits equals {year of course intake semester} subtract 1960. *Eg. For Semester 2021, first 2 digits = 2021 – 1960 = 61.*

Last 2 digits are generated base on the table below:

Course Term	Semester Month	Last 2 digits
Short Course	Jan	64
	Feb	65
	Mar	81
	Apr	21
	May	22
	Jun	23
	Jul	24
	Aug	25
	Sep	41
	Oct	61
	Nov	62
	Dec	63

5.2 ACAD_CAREER Mapping Rule

sTMS Course category (Code)	sTMS Course type (Code)	ACAD_CAREER value
Part-time Diploma (PTD)	Academic Full Qualification (FQ)	FTE
	Academic Certificate (MC)	FTE
	Academic Modular (SBMC)	FTE
Specialist Diploma (SD)	Academic Full Qualification (FQ)	PTD
	Academic Certificate (PDC)	PTD
	Academic Modular (SBMC)	PTD
Advanced Diploma (AD)	Academic Full Qualification (FQ)	ADSD
	Academic Certificate (PDC)	ADSD
	Academic Modular (SBMC)	ADSD
Diploma Conversion	Academic Full Qualification (FQ)	PTD
	Academic Certificate (PDC)	PTD
	Academic Modular (SBMC)	PTD
Certifiable Courses	SkillsFuture Series	CERT
	SkillsFuture Digital Confidence	CERT
	SkillsFuture Digital Workplace	CERT
	SGUnited Skills (SGUS) Programme	CERT
	Career Conversion Programme	CERT
	Work-Learn Bootcamp	CERT
	Certifiable Short Courses	SC
Non-Certifiable Courses	SkillsFuture Series	WSQ
	SGUnited Skills (SGUS) Programme	WSQ
	Work-Learn Bootcamp	SC
	Non-Certifiable Short Courses	SC
Work-Study Programmes	Work-Study Certificates Leading to Part-Time Diploma	FTE
	Work-Study Post-Diploma Specialist Diploma	PDP
	Work-Study Post-Diploma Leading to Specialist Diploma	PDP
	Work-Study Post-Diploma Advanced Diploma	PDP
	Workforce Skills Qualification (WSQ)	WSQ